Homework 1

Grant Talbert
January 30, 2025
PY 355 Section A1

Boston University

Problem 1

(a) The map $B \circ A = C : \mathbb{Z} \to \mathbb{N}$ is defined by

$$C(n) = B(n+1) = (n+1)^2 - n - 1 = n(n+1).$$

The image is $C(\mathbb{Z}) = \left\{ n^2 + n \in \mathbb{N} \mid n \in \mathbb{Z} \right\}$.

(b) The map A is a bijection. Let $n \in \mathbb{Z}$. Since $\mathbb{Z}$ is closed under addition/subtraction, we have $n - 1 \in \mathbb{Z}$, and thus $A(n-1) = n$. Therefore, A is surjective. Now suppose that $A(n) = A(m)$ for some $n, m \in \mathbb{Z}$. By definition of A,

$$A(n) = A(m) \implies n + 1 = m + 1 \implies n + 1 - 1 = m + 1 - 1 \implies n = m.$$

Therefore, A is injective, and thus bijective.

The map B is neither. Note that

$$B(2) = 4 - 2 = 2 = 1 + 1 = (-1)^2 - (-1) = B(-1).$$

Since $2 \neq -1$, B is not injective. Further, suppose $3 = n(n+1)$. We have

$$n^2 + n - 3 = 0 \implies n = \frac{-1 \pm \sqrt{1 + 12}}{2} = \frac{-1 \pm \sqrt{13}}{2} \notin \mathbb{Z}.$$

Therefore, there exists at least one element $3 \in \mathbb{N}$ such that $3 \notin B(\mathbb{Z})$, and thus B is not surjective.

Finally, note that since $A(\mathbb{Z}) = \mathbb{Z}$, we have that $B(\mathbb{Z}) = B(A(\mathbb{Z})) = C(\mathbb{Z})$, so C is not surjective because B is not surjective. Additionaly,

$$C(1) = 2 = (-1)^2 + (-1) = C(-1).$$

Therefore, C is not injective either.

Problem 2

1. $$\frac{\mathrm{d}}{\mathrm{d}x} 2^x = 2^x \ln 2.$$

2. $$\frac{\mathrm{d}}{\mathrm{d}x} \sinh x = \frac{\mathrm{d}}{\mathrm{d}x} \frac{e^x - e^{-x}}{2} = \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}x} e^x - e^{-x} = \frac{e^x + e^{-x}}{2} = \cosh x.$$

3. $$\frac{\mathrm{d}}{\mathrm{d}x} \cosh x = \frac{\mathrm{d}}{\mathrm{d}x} \frac{e^x + e^{-x}}{2} = \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}x} e^x + e^{-x} = \frac{e^x - e^{-x}}{2} = \sinh x.$$

4.
$$\frac{\mathrm{d}}{\mathrm{d}x}\tanh x = \frac{\mathrm{d}}{\mathrm{d}x}\frac{\sinh x}{\cosh x} = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = 1 - \tanh^2 x.$$

5.
$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\frac{x^2+2}{x^2-1} = \frac{\left(x^2-1\right)(2x) - \left(x^2+2\right)(2x)}{\left(x^2-1\right)^2}.$$

6.
$$\frac{\mathrm{d}}{\mathrm{d}x}x\ln x = \ln x + \frac{x}{x} = \ln x + 1.$$

7.
$$\frac{\mathrm{d}}{\mathrm{d}x}e^{-x^2} = e^{-x^2} * (-2x).$$

8.
$$\begin{aligned} & f(x) = x^x \\ \implies & \ln f(x) = x\ln x \\ \implies & \frac{\mathrm{d}}{\mathrm{d}x}\ln f(x) = \frac{\mathrm{d}}{\mathrm{d}x}x\ln x \\ \implies & \frac{f'(x)}{f(x)} = \ln x + 1 \\ \implies & f'(x) = \left(x^x\right)\left(\ln x + 1\right). \end{aligned}$$

Problem 3

The placement of the images got all messed up and I don't have time to fix it. Each figure's caption states what problem it belongs to. Sorry!

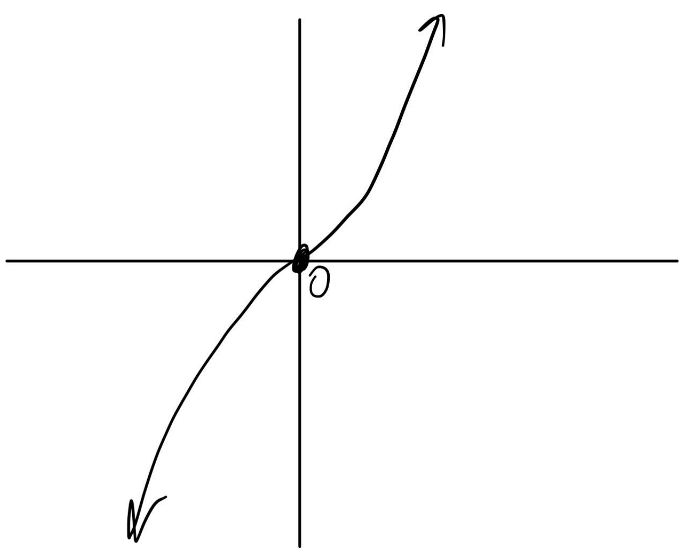

Figure 1: #3.1

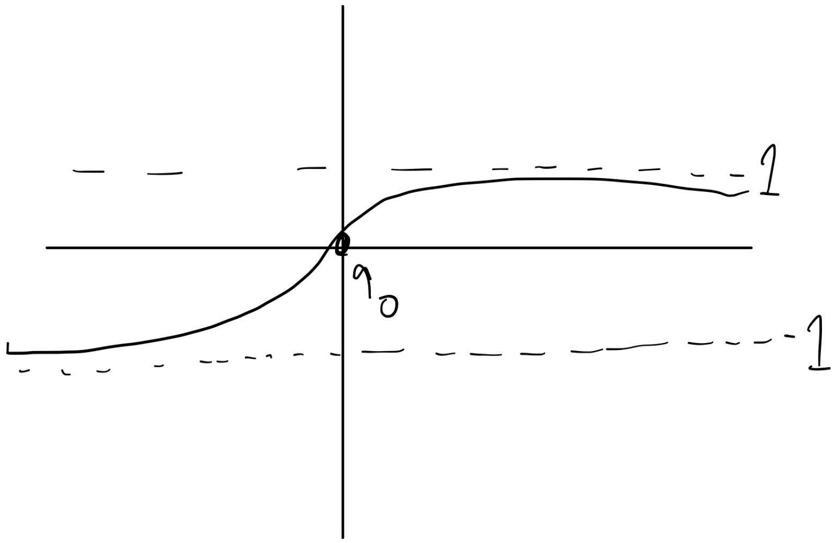

Figure 2: #3.2

Problem 4

1.
$$\frac{\mathrm{d}}{\mathrm{d}x}\sin^{-1}x = \frac{1}{\cos\left(\sin^{-1}x\right)} = \frac{1}{\sqrt{1-\sin^2\left(\sin^{-1}x\right)}} = \frac{1}{\sqrt{1-x^2}}.$$

2.
$$\frac{\mathrm{d}}{\mathrm{d}x}\cos^{-1}x = \frac{1}{-\sin\left(\cos^{-1}x\right)} = -\frac{1}{\sqrt{1-\cos^2\left(\cos^{-1}x\right)}} = -\frac{1}{\sqrt{1-x^2}}.$$

3.
$$\frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1}x = \frac{1}{\sec^2\left(\tan^{-1}x\right)} = \frac{1}{1+\tan^2\left(\tan^{-1}x\right)} = \frac{1}{1+x^2}.$$

Problem 5

$$f(0) = e^0 = 1.$$

$$f'(x) = e^{2x/(x+2)}\left(\frac{2(x+2)-2x}{(x+2)^2}\right) = e^{2x/(x+2)}\left(\frac{4}{(x+2)^2}\right).$$

$$f'(0) = e^0\left(\frac{4}{4}\right) = 1.$$

$$f''(x) = e^{2x/(x+2)}\left(\frac{16}{(x+2)^4}\right) + e^{2x/(x+2)}\left(\frac{-8}{(x+2)^3}\right).$$

$$f''(0) = e^0\left(\frac{16}{16}\right) + e^0\left(\frac{-8}{8}\right) = 0.$$

The Taylor expansion is

$$T_2(x) = 1 + x.$$

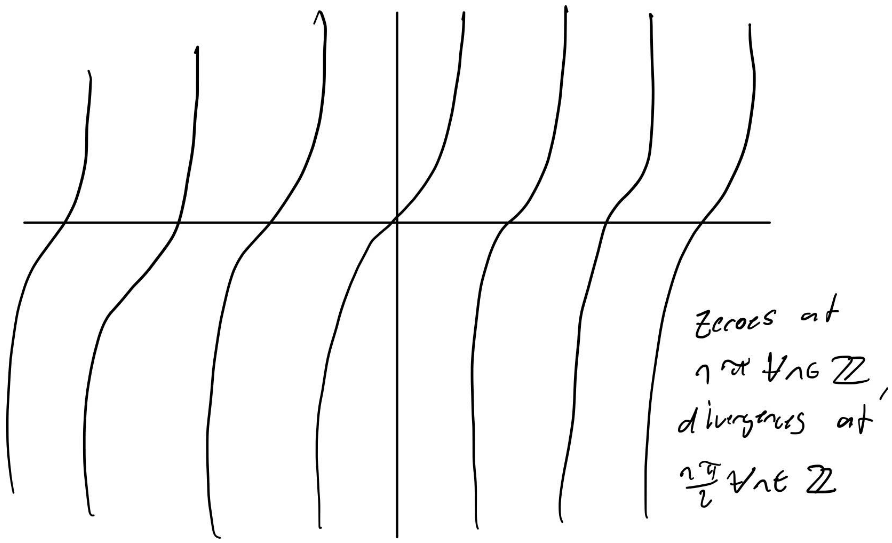

Figure 3: #3,3

Problem 6

1. $$\frac{\sin(0)}{0} = 0/0.$$

   Using L'Hopital's rule,
   $$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\cos x}{1} = \frac{\cos 0}{1} = 1.$$

2. $$\frac{a^2 + 2a - a^2 - 2a}{a^2 - a^2 - 2a + 2a} = 0/0.$$

   Using L'Hopital's rule,
   $$\lim_{x\to a} \frac{x^2 + (2-a)x - 2a}{x^2 - (a+2)x + 2a} = \lim_{x\to a} \frac{2x + 2 - a}{2x - a - 2} = \frac{2a + 2 - a}{2a - a - 2} = \frac{a+2}{a-2}.$$

3. $$\frac{e^{-0^4} - 1}{\left(e^{0^2} - 1\right)^2} = \frac{1-1}{(1-1)^2} = 0/0.$$

   Using L'Hopital's rule,
   $$\lim_{x\to 0} \frac{e^{-x^4} - 1}{\left(e^{x^2} - 1\right)^2} = \lim_{x\to 0} \frac{e^{-x^4}\left(-4x^3\right)}{2e^{x^2} \cdot 2x} = \lim_{x\to 0} \frac{-4x^2 e^{-x^4}}{2e^{x^2}} = \frac{0}{1} = 0.$$

4. $$\lim_{x\to\infty} x^n e^{-ax} = \lim_{x\to\infty} \frac{x^n}{e^{ax}} = \frac{\infty}{\infty}.$$

   Applying L'Hopital's rule,
   $$\lim_{x\to\infty} \frac{x^n}{e^{ax}} = \lim_{x\to\infty} \frac{nx^{n-1}}{ae^{ax}} \cdots \lim_{x\to\infty} \frac{(n \cdot (n-1) \cdot \dots \cdot (n - \lceil n \rceil))x^{n - \lceil n \rceil}}{a^{\lceil n \rceil} e^{ax}} = \frac{1}{\infty} = 0.$$

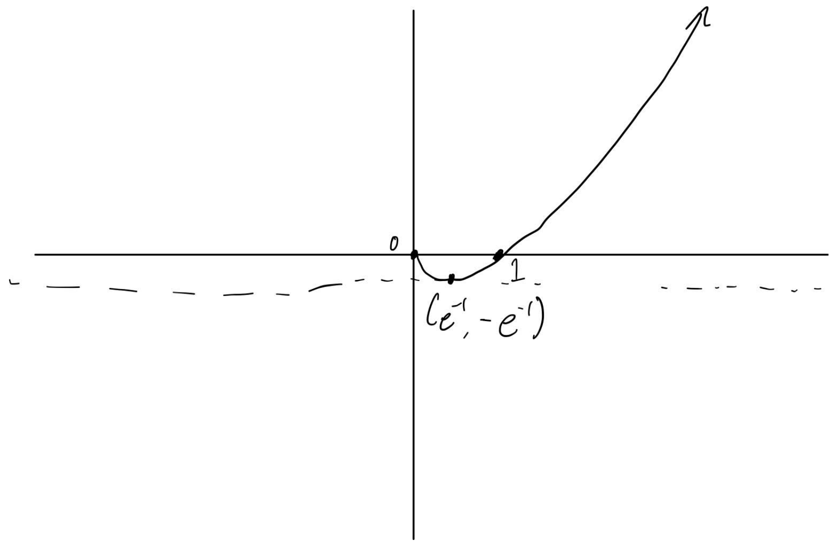

Figure 4: #3.4

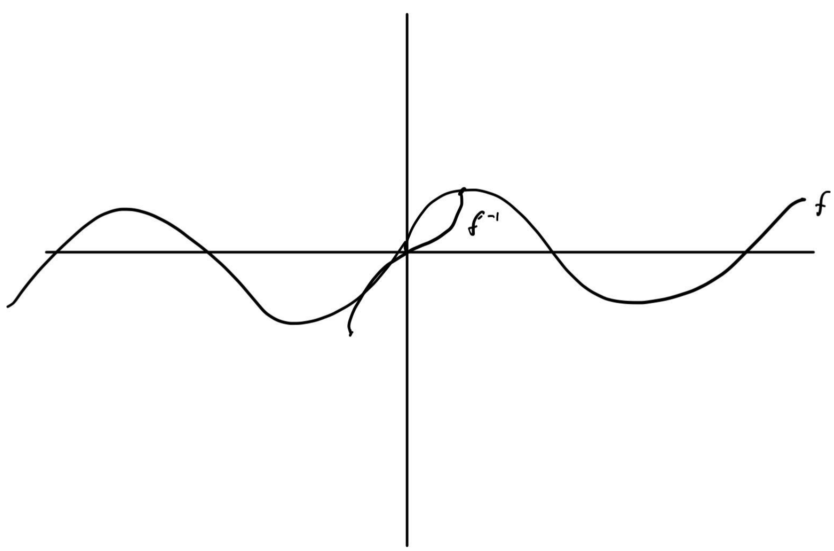

Figure 5: #4.1

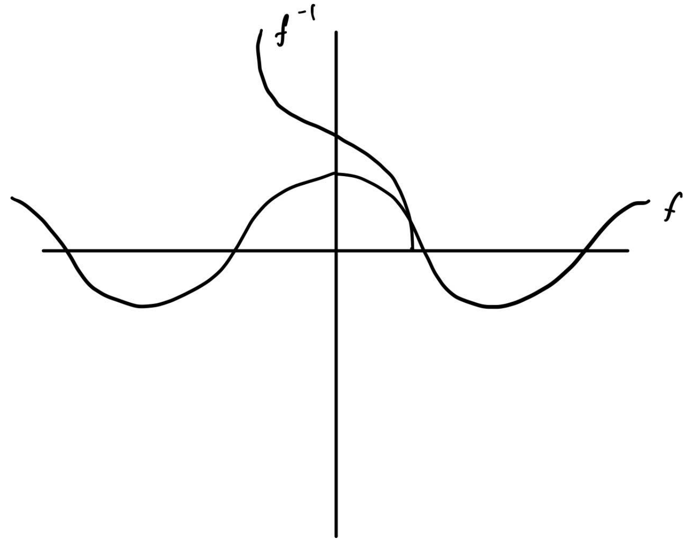

Figure 6: #4.2

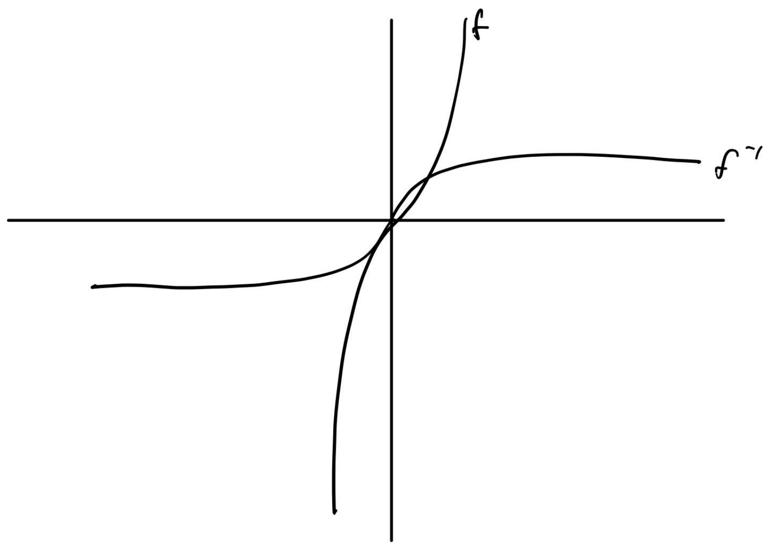

Figure 7: #4.3

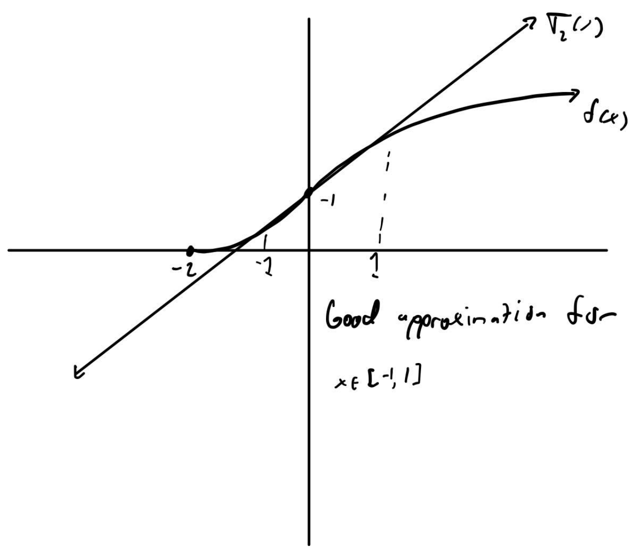

Figure 8: #5